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Palaeomagnetic rotations in opposite senses in southeastern Spain

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Abstract. 20 volcanic and 3 sedimentary sites of upper Miocene age were sampled in the Cabo de Gata region in southeastern Spain. Palaeomagnetic and rockmagnetic investigations yielded useful results in 15 volcanic and 1 sedimentary site. Clockwise and anticlockwise block rotations of adjacent blocks could be observed. The age of these rotations is younger than Tortonian. Deformation in the eastern part of the Betics seems to follow a different pattern than in the central and western part of the Subbetic. This behaviour can be related to the crustal structure in the eastern Betics.

Introduction

Earlier palaeomagnetic studies in the Betic Cordilleras (Osete et al., 1988; Platzman and Lowrie, 1992; Villalaín et al., 1992) have shown the occurrence of rotated palaeodeclinations in its central and western parts in sedimentary and volcanic rocks of Jurassic and Cretaceous age. Most of the rotations found in those studies are clockwise, with one exception, Sierra Gorda (Ogg et al., 1984; Osete et al., 1988; Platzman and Lowrie, 1992). This pattern has also been observed in magnetostratigraphic studies carried out in the same area (Ogg et al., 1984; Mazaud et al., 1986; Steiner et al., 1987). A more complex behaviour can be recognized in the eastern part of the Betic Cordilleras, where also very large rotations (Mazaud et al, 1986; Allerton et al, 1993) have been detected.

The Betic area has experienced a complex sum of movements from the Jurassic up to the present. The bulk of deformation seems to have ocurred since the Tertiary. The origin of the observed palaeomagnetic rotations is still being debated, as different events (middle Miocene thrusting or late Miocene/recent times strike-slip tectonics) could be responsible for them. There is still a lack of palaeomagnetic data which would allow to constrain the timing of rotation of palaeodeclinations. The present palaeomagnetic study was started in order to obtain more information about geographical distribution and timing of palaeomagnetic rotations in the Betics. For this purpose rocks of Neogene age, younger than those studied in other palaeomagnetic investigations in the Betic region were taken , mostly from volcanic but also from some sedimentary sites in the Cabo de Gata region, in southeastern Spain (fig.1).

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Geological setting

The Cabo de Gata calc-alkaline volcanic complex lies in the south-eastern tip of Spain (fig. 1). The main lithologies are dacites and andesites. Autoclastic breccias, often emitted as domes and seldom as lava flows, are very common. Also ash and pumice flows, as well as thin lapilli ash layers can be found (Fernández-Soler and Muñoz, 1988). Another outstanding feature

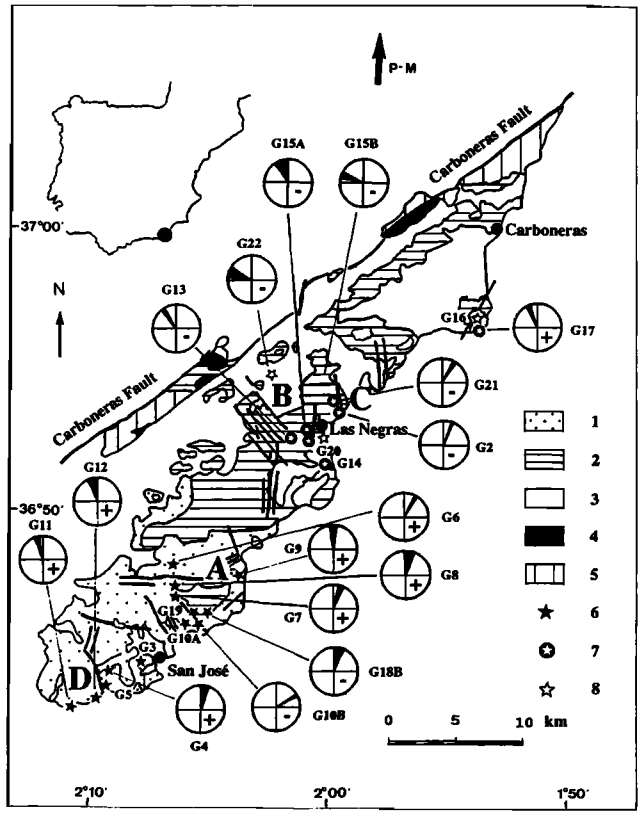


Figure 1. Schematic geological map and palaeomagnetic results of the Cabo de Gata region. 1: Volcanics, Lower Units; 2: Volcanics, Upper Units; 3: Miocene, Pliocene and Quaternary sedimentary rocks; 4: Internal Betics; 5: Volcanics, uncorrelated; 6: Sampling sites, Lower Units; 7: Sampling sites, Upper Units; 8: Sampling sites, sedimentary rocks. Circles show paleodeclinations with confidence limits (Demarest, 1983) and polarity (+: nomal, -: reversed). Different domains are indicated with capital letters (A,B,C,D). Thicker lines show the fault pattern in this area (from Montenat et al. 1990). A thick arrow shows the expected Pliocene-Miocene direction (Besse and Courtillot, 1991; Bógalo et al, in press).

is the development of hydrothermal systems during Miocene due to volcanic and subvolcanic activity. These systems are simultaneous with the emission of volcanic rocks, with fluids reaching temperatures of 400-500° (e.g. Fernández-Soler, 1992).

In order to describe calc-alkaline volcanism in the Cabo de Gata region, Fernández-Soler (1992) distinguishes two main volcano-stratigraphic units (fig.1). 13 sites were sampled in the oldest one (Lower Units), which is located in the southern part of the Cabo de Gata region, and 7 sites in the younger one (Upper Units), which can be mainly found north of the Lower Units, as well as in some isolated volcanic edifices (fig.1). The sampled sites were mostly domes or lava flows. K/Ar dating (Bellon et al., 1983; Di Battistini et al., 1987) yields ages between approximately 10 and 12/15 M.y. for the Lower Units and 8 M.y. for the Upper Units.

Marine sediments of Tortonian to Pliocene age lie on top of the Cabo de Gata volcanic units. Two sedimentary sites (oolithic limestones and fine-grained calcarenites) were sampled there. Another sedimentary site (bioclastic limestones) of Messinian age was sampled at the Níjar basin.

Sampling and laboratory methods

From each site 6-15 (in most cases 11 to 13) samples were cored with a portable drill, and one to four specimens were cut from each core. Cores were oriented in situ by means of a sun compass as well as a magnetic compass. Overlying sediments, local structural study and general bedding attitudes of sampling areas were used for bedding correction of volcanic formations, shallow dip values being obtained in most cases (tab. 1).

Table 1. Palaeomagnetic Results from Sites of the Cabo de Gata Region

Site/Unit	N	Dec	Inc	k	α_{95}	Rot	D	Bc	Ob
G3, L	9	220	-52	3	40			345,15	R
G4, L	11	12	55	54	6	9	A	320,15	N
G5, L	12	356	57	79	4			320,15	R
G6, L	12	28	33	103	4	25	A	13,38	Y
G7, L	11	20	54	89	5	17	A	350,25	N
G8, L	7	10	32	30	12	7	A	350,25	N
G9, L	11	2	60	70	6	-1	A	dip=0	N
G10A, L	6	252	-65	12	20			344,30	R
G10B, L	9	230	-51	333	3	47	A	344,30	N
G11, L	10	346	33	41	8	-17	D	dip=0	N
G12, L	10	345	43	38	8	-18	D	340,10	Y
G18B, L	9	195	-55	58	7	12	A	344,30	N
G19, L	12	189	-16	134	4			285,30	R
G2, U	12	201	-49	357	2	18	C	130,15	Y
G13, U	14	149	-48	67	5	-34	B	270,20	N
G14, U	10	8	-60	97	5			dip=0	R
G15A, U	4	160	-53	47	13	-23	B	5,20	Y
G15B, U	9	109	-42	21	10	-74	B	5,20	Y
G17, U	9	339	58	135	4	-24		dip=0	N
G21, U	9	213	-59	195	3	30	C	130,15	Y
G16, S	10	354	50	14	13			dip=0	R
G20, S	No consistent results							dip=0	R
G22, S	5	295	46	12	11	-68	B	90,10	Y

Site/Unit: Site number and unit to which each site belongs; L: Lower units; U: Upper units; S: Sedimentary rocks. N: Number of samples; Dec: Declination; Inc: Inclination; α_{95}: Radius of 95% confidence circle; k : Precision parameter; Rot: Rotation of palaeodeclinations with respect to expected direction (see text); Positive values are for clockwise and negative values for counterclockwise rotations; D: Domain to which each site belongs. BC: Bedding correction (strike and dip). Ob (observations): Y: Bedding correction applied; N: ChRM acquired after dipping of bedding plane (bedding correction not applied); R: Rejected site.

Remanent magnetization of all volcanic units was measured with a Molspin spinner magnetometer and some of the sedimentary samples had to be measured with a cryogenic magnetometer because of their weak magnetization. From each site four pilot samples were chosen, two of them being subjected to stepwise thermal demagnetization up to 700°C and two of them to stepwise AF cleaning up to 100 mT peak value. According to the results of the preliminary experiments, all other samples were subjected to less detailed thermal or AF demagnetization. During thermal demagnetization initial susceptibility at room temperature was measured after each step. Directions of magnetization components were calculated using linear regression.

In order to identify the magnetic minerals responsible for remanent magnetization and to obtain information about their palaeomagnetic stability, several rockmagnetic experiments were carried out, including determination of thermomagnetic curves and hysteresis parameters, measurement of IRM acquisition curves and thermal demagnetization of two component IRM experiments, ore microscopy and measurement of anisotropy of magnetic susceptibilty (Calvo, 1992).

Results

Calk alcaline rocks of Cabo de Gata

Demagnetization vector plots of the rocks belonging to calc-alkaline volcanics of the Cabo de Gata region displayed some differences between sites, and in some cases also within-site differences could be recognized. Nevertheless, only five sites had to be rejected (tab.1), in two cases because of an unreasonable high within-site scatter (G3 and G10A), in one case because its remanence showed a viscous present-day field direction (G5), in one case because of the superposition of two components of equal coercivity and unblocking temperature spectra (G19), and in another case because an unreasonable high palaeodeclination was observed (G14).

In four cases, a single-component behaviour could be observed, magnetization being carried by magnetite, as shown by unblocking temperatures during thermal demagnetization (fig. 2a) and thermomagnetic curves. In several cases, two more or less superimposed components could be distinguished, one of them being carried by titanomaghemite with a low unblocking temperature spectrum and often with very low coercivities (fig. 2b). Characteristic remanence is carried by magnetite or low-Ti titanomagnetite, although the presence of titanohematite could be recognized in several cases. These titanohematites were produced by hydrothermal alteration (Ade-Hall et al., 1971), as could be seen by means of ore-microscopic studies on polished specimens. They do not seem to carry a different palaeomagnetic component, since the hydrothermal alteration suffered by the studied region was simultaneous with the emission of volcanic rocks (e.g. Fernández-Soler, 1992). Sometimes, high coercivities -in one case MDF being greater then 80 mT- could be observed. Low-Ti titanohematites or subdivision of titanomagnetites due to granulation during hydrothermal alteration could be responsible for high coercivities (e.g. Løvlie, 1988).

Sedimentary sites in the Cabo de Gata region

In the site at the Nijar basin (G22) a useful result was obtained, the carrier of remanence being hematite, although an un-

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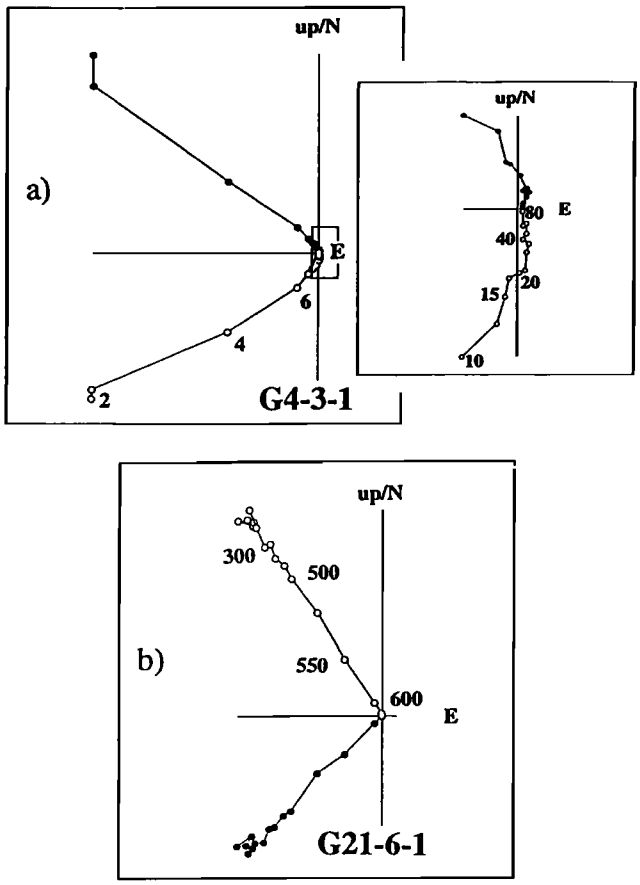


Figure 2. Representative vector demagnetization plots from specimens of the Cabo de Gata region. Solid circles are for the horizontal projection and open circles for the vertical projection in the E-W vertical plane. a) AF-demagnetization, demagnetization steps being indicated in mT; b) Thermal demagnetization, demagnetization steps being indicated in °C.

stable low-coercivity component, carried by (titano)magnetite or (titano)maghemite could be observed. The two other sedimentary sites sampled in this area (G16 and G20) yielded no useful results, weak magnetization and the presence of a VRM being the reason for this behaviour.

Discussion and conclusions

After comparing the palaeomagnetic results obtained in each site with the expected Pliocene/Miocene direction (D=2.9°, I=49.3°, calculated from poles from Besse and Courtillot (1991) and Bógalo et al., in press), we can recognize several sites with homogeneously rotated palaeodeclinations (fig.1 and tab. 1), so that different domains can be distinguished: A group of mostly clockwise rotated directions, which we shall call from now on domain A north of San José and another group of counterclockwise rotated directions (domain B) east of Las Negras are easily recognized. The latter ones lie close to two sites rotated clockwise (domain C), both sectors being separated by a fault. On the southern tip of Cabo de Gata two counterclockwise rotated directions -G11 and G12 (domain D?)- can be found. The mean directions for domains A and B are shown on tab. 2.

The question can be raised if secular variation alone could be responsible for the observed rotations. In the case of domains A and B this is highly unlikely. In one case six homogeneously rotated sites guarantee the significance of the rotation while in the other case a sedimentary site (G22) confirms the sense of rotation of the volcanic sites. The fact that sites G2 and G21 at las Negras are rotated in the opposite direction than G13, G15A, G15B and G22, both areas being separated by faults, points to a tectonic origin of these rotations. Site G15A and G15B belong to the same unit, although the former shows greater alteration. The fact that an almost N-S trending fracture could be observed between both sites, drove us to consider them separately.

For the following reasons we consider ChRM as being of primary origin: a) Normal and reversed directions were detected; b) In many sites, particularly in those belonging to the oldest volcanic rocks, a considerable part of NRM, carried by minerals of low unblocking temperature and coercivity, shows a present-day field direction, which is different from the ChRM found in the same rocks, which is carried by stable minerals; c) Hydrothermal alteration (maximum temperatures of 400-500°), which affects many of the studied sites, is simultaneous with volcanism (Fernández-Soler, 1992); d) ChRM is carried by stable magnetite or low-Ti titanomagnetite and, in some cases, also by high-coercivity titanohematite with high MDF; e) Specifically, the younger units of Cabo de Gata show in most cases a ChRM direction carried by magnetite produced by deuteric oxidation and with reversed polarity.

Small rotated areas can be therefore found in southeastern Spain. An outstanding feature of these rotations is that its pattern is not homogeneous, and adjacent areas appear to have rotated in opposite senses. Palaeomagnetic investigations carried out in Israel (Ron et al., 1984) or in western Anatolia (Kissel et al., 1987) have shown the occurrence of block rotations by different amounts and in opposite senses.

Garfunkel and Ron (1985) propose block rotation models in areas with simultaneous deformation of adjacent domains of conjugate faults. The sense of block rotation and fault movements depends on the orientation of fault domains with respect to overall stress, but rotations and shears in opposite senses can compensate for each other on a regional scale. Cobbold et al. (1989) carried out a series of experiments, in which models of continental litosphere were subjected to different stress regimes. The result of one of those experiments, in which coaxial stretching was combined with a small amount of left-lateral simple shearing, showed the development of two approximately equal conjugate fault sets, left lateral faults being longer than right-lateral faults, which were confined in a central domain. Block rotations in opposite senses, with right lateral domains showing counterclockwise and left-lateral domains clockwise rotations, could be observed. This pattern resembles the structural setting in southeastern Spain, with longer NE-SW and N-S trending left-lateral faults and shorter NW-SE trending right-lateral

Table 2. Mean Directions for Rotated Domains

Dom	N	Dec	Inc	K	α_{95}	Rot
A	7	20	50	29	11	17
B	4	312	49	22	20	-51

Dom: Domain; N: Number of samples; Dec: Declination; Inc: Inclination; k: precision parameter; α_{95}: Radius of 95% circle of confidence; Rot: Rotation of domains with respect to expected direction (see text); Positive values for clockwise and negative values for counterclockwise rotations;

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faults (e.g. Montenat et al., 1987). Thus, both models propose structural settings in which such kinds of rotations are possible.

While palaeomagnetic results obtained in the central and western part of the Betic Cordilleras (Osete et al., 1988; Platzman and Lowrie, 1992; Villalaín et al., 1992) show the occurrence of homogeneous clockwise rotations, the structural pattern of this region showing a predominance of short NNW-SSE trending faults, the results of the present palaeomagnetic study show that the deformation pattern of the eastern Betics, where long NE-SW trending left-lateral faults are the main structures, is different. This seems to point to a causal relation between crustal structure and surface deformation, particularly if we take into account that the region located at the east of the NE-SW trending left-lateral fault zone in the eastern Betics seems to consist of thinned crust (Banda and Ansorge, 1980; Banda et al., 1993).

The age of rotations is younger than 11-12 M.y. (Tortonian), which is the age of the volcanic rocks studied in the Cabo de Gata region. The activity of the NE-SW trending faults is very recent and postdates the main volcanic activity. The magnetic polarity of palaeodirections matches the chronological zoning of volcanism in this region (G10B and G18B belong to the Lower Units and have reversed polarity, but lie close to the boundary between Lower and Upper Units) thus confirming it on an independent way. This agrees with a northeastward migration of volcanism in this area.

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